

Trained PING streams sequential digits without retraining

exp048 · 8 June 2026

Abstract

A PING network trained one digit at a time on 200 ms trials can classify a *stream* of digits at a fraction of that time each, with no retraining: only the readout’s integration window changes. At $\tau = 50$ ms (≈ 2 gamma cycles) 4 of the 5 sequential MNIST digits classify correctly, the readout flipping within one cycle of each transition. A 2-D sweep over (τ , input rate) reveals a sub-cycle failure floor below ≈ 15 ms and a broad high-accuracy plateau above it. An extended 200 ms rate slice remains at chance through 0.5 Hz and becomes clearly informative by 2 Hz, locating the encoder’s low-rate information floor.

Method

Canonical exp025 PING baselines (1,024 excitatory + 256 inhibitory cells, trained one MNIST digit per 200 ms trial, three seeds 42, 43, 44). Everything here is **inference-only** at the trained $\Delta t = 0.10$ ms; the weights are never updated. The 2-D sweep averages over the three seeds; the single-stream demos use seed 42.

A stream of digits, each shown for τ ms, is classified in one forward pass. The *only* change from training is a sliding readout window:

1. **Encode** each digit as a Poisson spike train over τ ms across 784 input channels, one per pixel, firing rate proportional to pixel intensity at the chosen input rate.
2. **Concatenate** the per-digit trains into one input stream. Transitions are *hard switches*: the Poisson rate flips instantaneously at $t = k\tau$, with no blending.
3. **One forward pass** of the trained network over the whole stream (no retraining), at the trained Δt .
4. **Integrate evidence** in a non-spiking output leaky-integrator, one unit per class:

$$v_{\text{out}}(t) = \beta_{\text{out}} v_{\text{out}}(t-1) + \frac{1 - \beta_{\text{out}}}{\Delta t} \mathbf{s}^E(t-1) W_{\text{out}}.$$

5. **Read a sliding window.** Average v_{out} over the trailing τ -window and softmax it:

$$\text{logits}(t) = \frac{\Delta t}{\tau} \sum_{u=t-w+1}^t v_{\text{out}}(u), \quad p(\text{class}, t) = \text{softmax}(\text{logits}(t)),$$

At training the average ran over the *whole* trial; the trailing window is the single change.

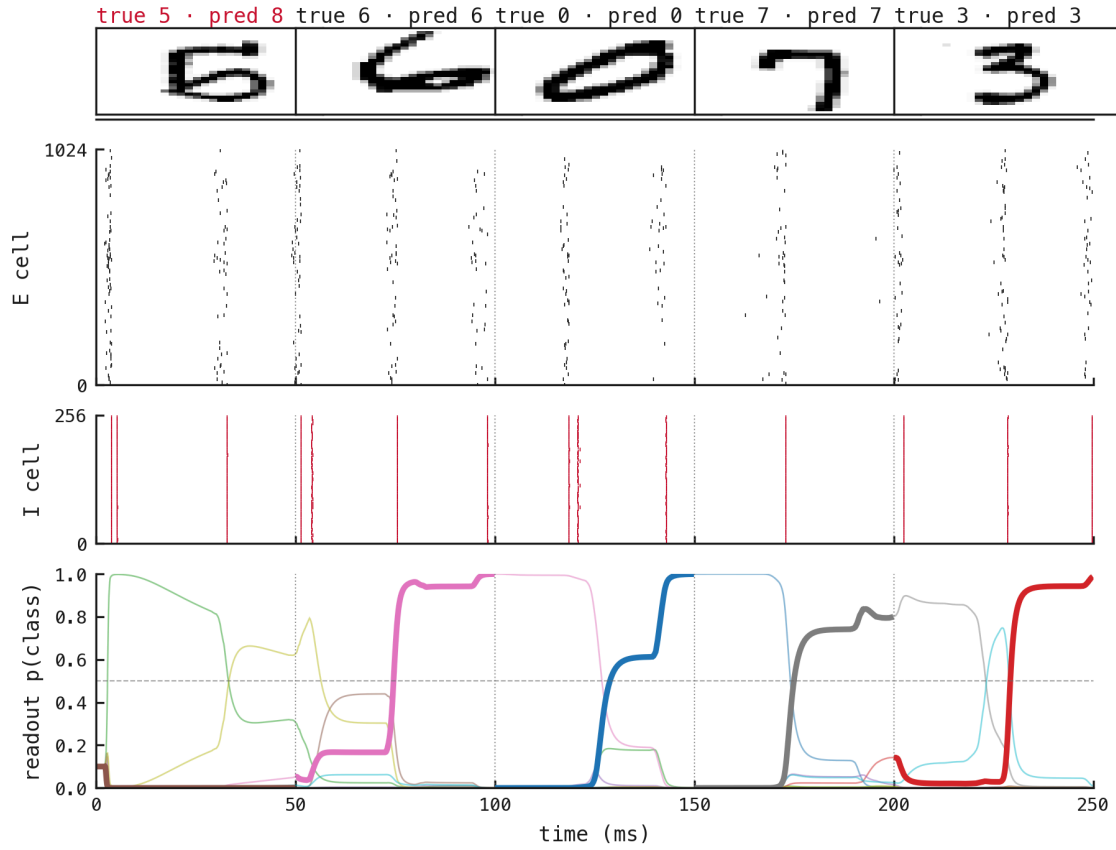
6. **Predict per segment** as $\arg \max_c p(\text{class} = c, t)$, read at the end of each digit’s τ -window.

Here $\mathbf{s}^E(t)$ is the excitatory spike vector, W_{out} the trained readout (unchanged), and $\beta_{\text{out}} = \exp(-\Delta t / \tau_{\text{out}})$ the output leak with $\tau_{\text{out}} = 2$ ms. The probability trace $p(\text{class}, t)$, the network’s online confidence in each digit, is what the figures plot.

The grid sweeps τ over 10, 15, 25, 40, 50, 75, 100, 200 ms and input rate over 5, 10, 25, 50, 100, 200 Hz per channel, giving 8×6 cells, each 40 streams $\times$ 10 digits $\times$ 3 seeds (1200 segments per cell). It extends down to $\tau = 10$ ms (≈ 0.36 of a gamma cycle) to resolve the sub-cycle regime.

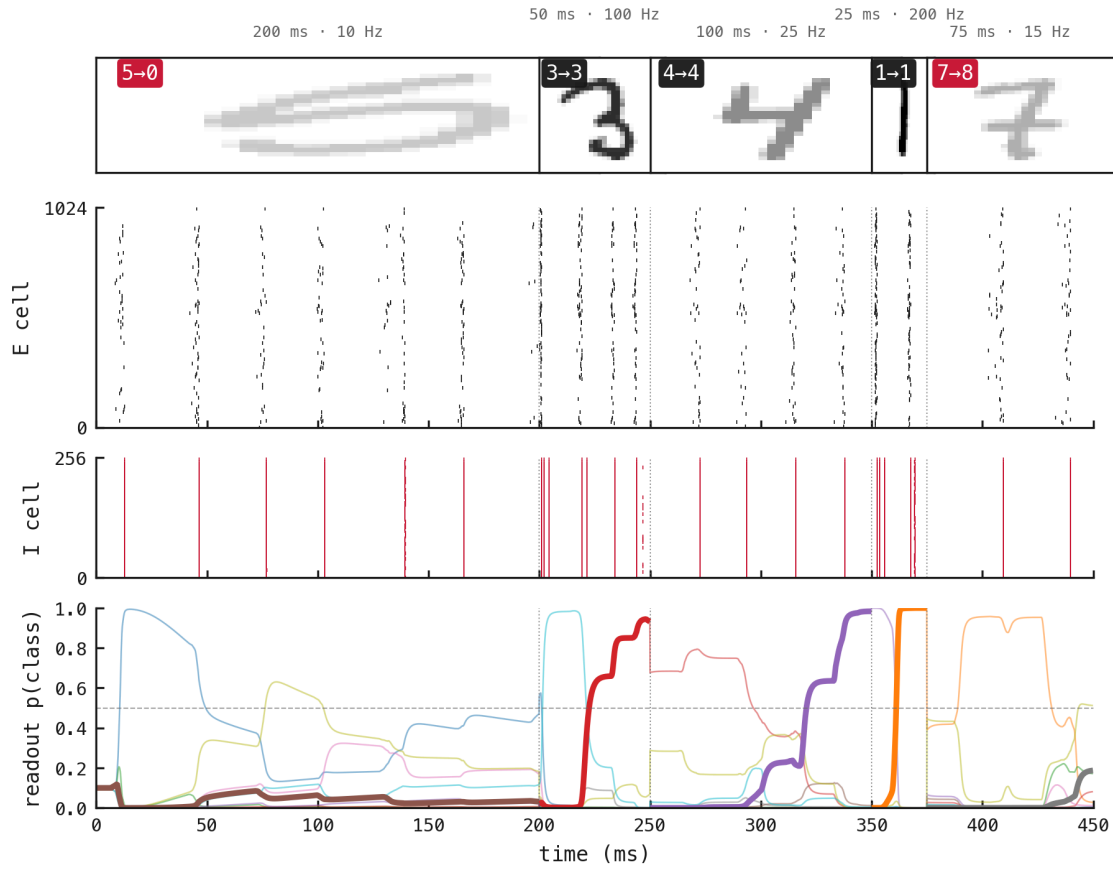
To resolve the encoding-rate floor below the grid, an additional 200 ms slice sweeps rates from 0.01 to 3 Hz. Each cell contains 10 streams of 10 digits across the same three trained seeds. The published 5–200 Hz points are the matching 200 ms row of the grid.

Results



replot

Figure 1: Five sequential digits (5, 6, 0, 7, 3) at $\tau = 50$ ms each (≈ 2 gamma cycles per digit, 250 ms total, 25 Hz input). The gamma cadence (≈ 28 ms) is preserved across the stream; the readout flips to the new digit within one cycle of each transition and reaches near 1.0 by the segment's end. **4/5 correct** (the leading 5 is misread). The figure makes four things visible at once: sparse E firing, the I-burst clock, class-tracking readout, and re-identification within ≈ 1 cycle of each switch.



exp048-replot

Figure 2: A harder test: each digit gets its *own* duration and input rate within one stream (durations 25–200 ms, rates 10–200 Hz). Thumbnail opacity $\propto$ input rate (faint = weak drive, bold = strong); the sliding window uses each segment's own τ , so every digit's prediction respects its presentation window. **3/5 correct** here: the two errors fall on the weakest-drive segments (a 5 at 10 Hz and a 7 at 15 Hz), the two lowest input rates in the stream.

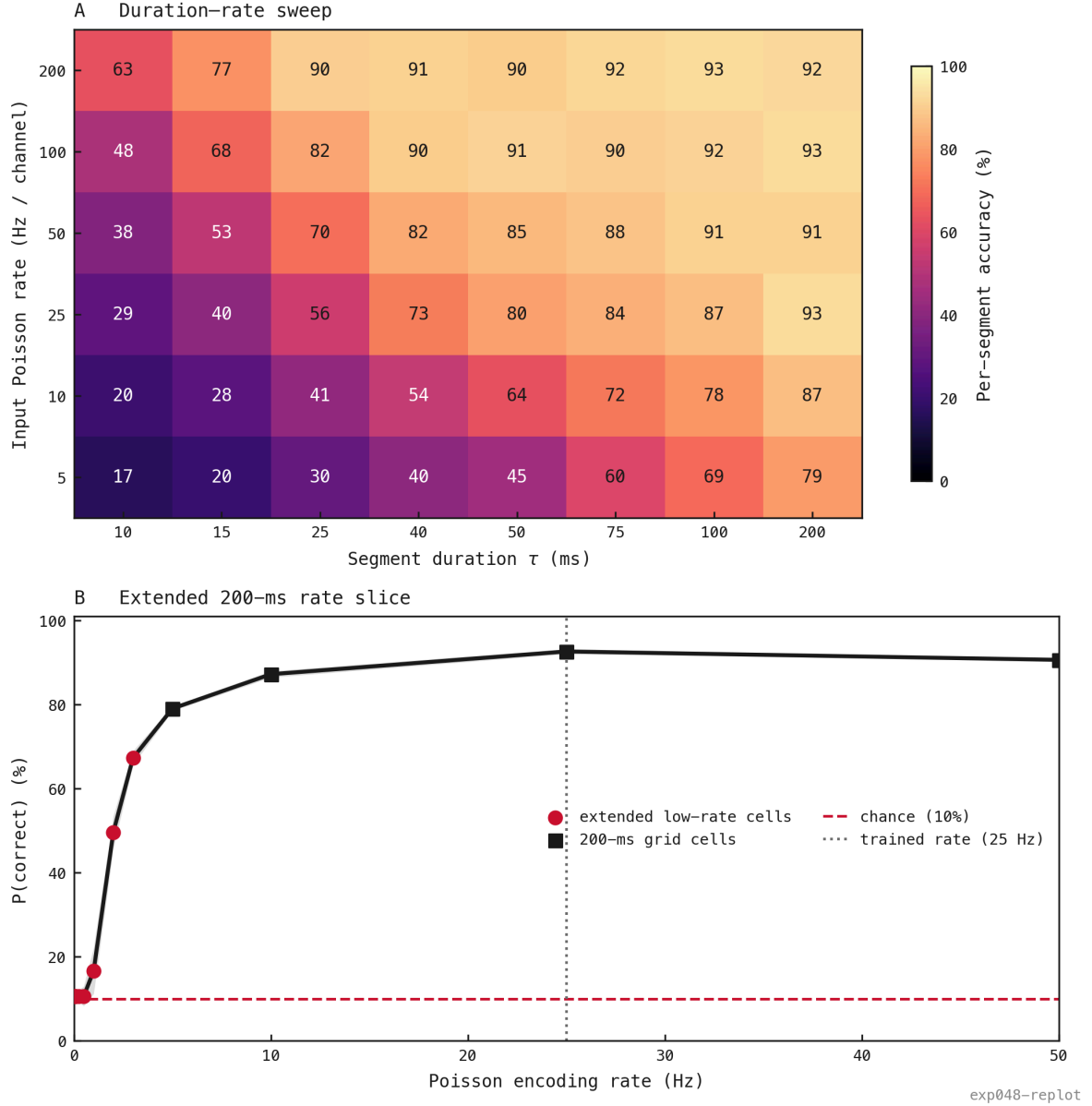


Figure 3: Temporal and encoding-rate limits of the frozen PING classifier. **(A)** Per-segment accuracy over the full duration $\times$ rate grid (3 seeds $\times$ 1200 segments per cell). The sub-cycle regime fails even under strong drive, while above one cycle duration and rate trade off along diagonal iso-accuracy contours. **(B)** The 200 ms row extended below 5 Hz on a linear rate axis. Circles are the new low-rate cells; squares come from panel A. The dashed line marks ten-class chance and the dotted line the 25 Hz training rate. Accuracy remains on its empty-input floor through 0.5 Hz, is clearly informative by 2 Hz, and reaches 79.1% at 5 Hz.

The extended slice distinguishes a nonviable encoder regime from ordinary classification errors under weak evidence. In the variable-condition stream, the failed 5 received 10 Hz for 200 ms, yet that condition reaches 87.3% across the population. Its error is therefore natural trial-level variation, not evidence that 10 Hz is intrinsically too low. The failed 7 at 15 Hz and 75 ms is likewise above the empty-input rate floor, although its shorter window supplies less total evidence. Rates below 0.5 Hz are not useful operating points; 2 Hz is the lowest clearly informative tested rate and 5 Hz is a practical lower bound for future sweeps.